



Problem solving with theoretical probability

- 1 The table shows the possible outcomes when two six-sided dice are rolled and their numbers are summed. Which outcome is most likely to happen? Tick **1** correct answer

		dice 1					
		1	2	3	4	5	6
dice 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

☐ 5
☐ 6
☒ 7
☐ 8
☐ 9

- 2 The table shows the possible outcomes when two six-sided dice are rolled and their numbers are summed. Which outcomes have a probability of $\frac{3}{36}$? Tick **2** correct answers

		dice 1					
		1	2	3	4	5	6
dice 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

☐ 2
☒ 4
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- 3 The table shows the possible outcomes when two six-sided dice are rolled and their numbers are summed. What is the probability that the dice sum to 5? Tick **1** correct answer

		dice 1					
		1	2	3	4	5	6
dice 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

☐ $\frac{1}{6}$

☐ $\frac{5}{6}$

☐ $\frac{4}{15}$

☒ $\frac{4}{36}$

☐ $\frac{5}{36}$

- 4 Which calculation would increase 80 by 50%? Tick **1** correct answer

- ☐ 80×0.5
- ☒ 80×1.5
- ☐ 80×50
- ☐ 80×150

- 5 Increase 20 by 40%. Fill in the blank

28

- 6 What is 1.5% of 1000? Fill in the blank

15

